

Excellent!

$$\frac{9}{9} \Rightarrow 100\%$$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	10
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Name: _____
These questions assess your understanding of the material from Chapters 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. When Stephen gets in his car, it lowers by 3.0 cm. If Stephen's mass is 63. kg, then what is the force constant of his car's suspension system?

ANSWER: $2.06 \times 10^4 \text{ N/m}$ ✓

$$\begin{aligned} \uparrow -k\Delta x \\ \downarrow mg \\ -mg = -k\Delta x \\ \Rightarrow \frac{mg}{\Delta x} = k = \frac{(63 \text{ kg})(9.8 \text{ m/s}^2)}{0.03 \text{ m}} \\ = 20580 \text{ N/m} \end{aligned}$$

2. The spring of a toy gun has a force constant of $66.2 \frac{\text{N}}{\text{m}}$ and is compressed 23.9 cm. Neglecting friction and the mass of the spring, at what speed will a 1.48-g projectile be ejected from the toy gun?

ANSWER: 50.547 m/s ✓

$$\begin{aligned} PE &= KE_f \\ \frac{1}{2} kx^2 &= \frac{1}{2} m v_f^2 \Rightarrow \sqrt{\frac{kx^2}{m}} = v_f \\ \Rightarrow \sqrt{\frac{(66.2 \frac{\text{N}}{\text{m}})(0.239 \text{ m})^2}{(1.48 \times 10^{-3} \text{ kg})}} &= 50.547 \text{ m/s} \end{aligned}$$

3. Suppose you have a tube that is closed at one end. Its length is 21. cm and the air temperature is 1.4°C . What is the frequency of its first overtone? $n = 3$

$$v_w = (331) \sqrt{\frac{273 + 1.4}{273}} \quad L = 0.21 \text{ m} \quad \text{ANSWER: } 1185.17 \text{ Hz} \quad \checkmark$$

$$v_w = 331.848 \text{ m/s}$$

$$\Rightarrow f_3 = 3 \frac{v_w}{4L} = 3 \frac{331.848}{4(0.21)} = 1185.17 \text{ Hz}$$

+3

4. Suppose that a 1.89×10^3 -kg car has a suspension system with a force constant $1.09 \times 10^4 \frac{\text{N}}{\text{m}}$. If the car hits a bump and bounces with an amplitude of 16. cm, then what is its maximum vertical velocity assuming no damping occurs?

$$v(t) = \sqrt{\frac{k}{m}} X \sin(90)$$

ANSWER: 0.384 m/s

$$v(t) = \sqrt{\frac{(1.09 \times 10^4)}{(1.89 \times 10^3)}} (0.16) = 0.384 \text{ m/s}$$

5. A medical imaging device produces ultrasound by oscillating with a period of $0.324 \mu\text{s}$. What is the frequency of this oscillation?

$$f = \frac{1}{T} = \frac{1}{0.324 \times 10^{-6} \text{ s}}$$

ANSWER: $3.086 \times 10^6 \text{ Hz}$

$$= 3.086 \times 10^6 \text{ Hz}$$

6. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be 121. Hz and the air temperature is $25.^\circ\text{C}$, then what should the length of the tube be?

$$f_n = v \frac{1}{4L} \Rightarrow L = \frac{v}{4f_n}$$

ANSWER: 71.45 cm

$$v_w = (331) \sqrt{\frac{273+25}{273}} = 345.8237 \text{ m/s}$$

$$\Rightarrow L = \frac{(345.8237)}{4(121 \text{ Hz})} = 0.7145 \text{ m or } 71.45 \text{ cm}$$

7. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the frequency of oscillations for a car with shoddy shock absorbers if the car's total mass is $2.14 \times 10^3 \text{ kg}$ and the force constant of the suspension system is $2.77 \times 10^4 \frac{\text{N}}{\text{m}}$.

$$T_{SHO} = 2\pi \sqrt{\frac{m}{k}}$$

ANSWER: 0.5726 Hz

$$\Rightarrow 2\pi \sqrt{\frac{2.14 \times 10^3}{2.77 \times 10^4}} = 1.74641 \text{ s}$$

$$\Rightarrow f = \frac{1}{1.74641} = 0.5726 \text{ Hz}$$

+4

8. Suppose a 0.304-kg object is connected to a horizontal spring. There is friction between the object and the surface, and the coefficient of kinetic friction μ_k is 0.119. What total distance does the object travel if it is released from rest at a distance of 35. cm from its equilibrium position? The force constant of the spring is $37.0 \frac{\text{N}}{\text{m}}$.

ANSWER: 6.39 m

$$E = f \cdot d$$

$$\Rightarrow d = \frac{E}{f}$$

$$E = \frac{1}{2} k x^2$$

$$f = \mu_k m g$$

$$d = \frac{(\frac{1}{2} k x^2)}{\mu_k m g} = \frac{(\frac{1}{2})(37)(0.35)^2}{(0.119)(0.304)(9.8)} = 6.39 \text{ m}$$

9. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of $816 \frac{\text{kg}}{\text{m}^3}$ to muscle tissue with a density of $1.125 \times 10^3 \frac{\text{kg}}{\text{m}^3}$. The speed of ultrasound in the fat is $1.41 \times 10^3 \frac{\text{m}}{\text{s}}$ and the speed of ultrasound in the muscle tissue is $1.61 \times 10^3 \frac{\text{m}}{\text{s}}$.

ANSWER: 4.976 %

$$a = \left(\frac{Z_2 - Z_1}{Z_1 + Z_2} \right)^2$$

$$Z_2 = (1.125 \times 10^3)(1.61 \times 10^3) = 1811250$$

$$Z_1 = (816)(1.41 \times 10^3) = 1150560$$

$$a = \left(\frac{1811250 - 1150560}{1811250 + 1150560} \right)^2 \times 100 = 4.976 \%$$

+2

1. $k = 2.06 \times 10^4 \frac{\text{N}}{\text{m}} \therefore \vec{F} = -k\Delta\vec{x}, k = -\frac{\vec{F}}{\Delta\vec{x}}$
2. $v = 50.5 \frac{\text{m}}{\text{s}} \therefore KE = PE_{elastic}$
3. $f = 1.19 \times 10^3 \text{ Hz} \therefore f_n = n\frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273\text{K}}}$
4. $v_{max} = 0.384 \frac{\text{m}}{\text{s}} \therefore v(t) = \sqrt{\frac{k}{m}} X \sin \frac{2\pi t}{T}$ and the max of sine's range is 1.
5. $f = 3.09 \times 10^6 \text{ Hz} \therefore T = \frac{1}{f}$
6. $L = 0.7145 \text{ m} \therefore f_n = n\frac{v_w}{4L}, v_w = \left(331 \frac{\text{m}}{\text{s}}\right) \sqrt{\frac{T}{273\text{K}}}$
7. $f = 0.573 \text{ Hz} \therefore T_{SHO} = 2\pi\sqrt{\frac{m}{k}}, f = \frac{1}{T}$
8. $d = 6.39 \text{ m} \therefore \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k mg, PE = \frac{1}{2} kx^2$
9. $a = 4.98\% \therefore Z = \rho v, a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$